

The Effects of Brownfield Designation on Parcel Tax Valuation in Worcester, MA

Introduction: Background, Research Questions, and Objectives

The US Environmental Protection Agency created the Brownfields Program in 1995 as a systematic way of addressing hazardous material and other pollutant contaminations in the United States. Defined by the Small Business Liability Relief and Brownfields Revitalization Act of 2002, a brownfield is a “real property, the expansion, redevelopment, or reuse of which may be complicated by the presence or potential presence of a hazardous substance, pollutant, or contaminant.” (Gillmor, 2001). There is general consensus in the academic literature regarding the negative impact brownfields have on the properties they’re located within as well as properties within varying proximities (De Sousa et al., 2009). Simply put, brownfields decrease the value and tax generating ability of a property (and properties within varying proximities) as well as limit or prevent the development or reuse of these properties. As of March 16, 2026, Worcester has 47,409 parcels and 1,715 brownfield sites.

We were interested in understanding the specific relationships between brownfields and property tax valuation. The driving question for this project centers on the effects of brownfield designation on parcel tax valuation and, if such effects are identified, the nature of those effects. For this, we set the total valuation as our response variable (Y) and list price, lot size, number of RTNs within 500ft, duration days, inactive days, and contaminant type as our predictor variables (Xs). Our objective is to answer the following questions: 1) Do brownfields affect parcel tax valuation? 2) If there is a relationship, what trends exist between brownfield sites and parcel valuation? 3) What do the model diagnostics show regarding the linear fit of our model? 4) Is there a significant spatial relationship between response and predictor, or is tax valuation significantly spatially autocorrelated?

Data Collection and Pre-Processing

To complete this analysis, we draw from two main data sources. The Waste Site and Reportable Releases dataset was filtered to Worcester and downloaded from the Energy and Environmental Affairs Data Portal (MassDEP) on March 16, 2026. This dataset includes information on reported brownfields throughout Worcester such as release tracking number (RTN), regulatory status and date, and contamination type. The property tax parcel GIS dataset was accessed from MassGIS through the ArcGIS Online portal and filtered to Worcester. This dataset includes information about each parcel such as list price (last sale price), lot size, and total valuation.

The pre-processing of this data mostly took place on ArcGIS Pro. First, we geocoded the brownfields dataset using the ArcGIS Online geocoder. Next, brownfield points that were located outside of Worcester were removed and points that were outside a parcel boundary were moved to the nearest polygon (these included points that took place on the interstate highway, roadways, ponds/lakes, etc.). Additional fields were also added to this feature class

including Duration Days, Inactive Days, and Contamination Types (categorical data). Next, a spatial join of Worcester property tax parcels and brownfields within 500ft created a new output of the combination between parcels and brownfields. A new additional field was also created during this join to count the RTN joins per parcel. In order to ensure this dataset was ready for analysis, a quality control assessment was run; we removed railroad parcels that spanned the entire city and had hundreds of RTN joins (much higher than any other parcels), removed parcels with total value of 0 or null, and all other null values in the data set for list sale, RTN joins, duration days, and inactive days were given a 0 value as well. This process ensures that the entire dataset can be used during the analysis in R. Finally, the dataset was exported from ArcGIS Pro as a CSV file and brought into R for the rest of the analysis.

Methods: Spatial Autocorrelation

Once the table has been processed in GIS, we wanted to determine if our response variable, tax valuation (dollars), is spatially autocorrelated. To do this, we removed outliers and parcels with a tax valuation of \$0. High value parcels were removed by using the interquartile range (IQR) method. First, we computed the quartiles for this dataset in R. We then calculated the IQR ($IQR = Q3 - Q1$), and then fed the range into the formula for the upper fence ($Q3 + (3 \times IQR)$). All parcels with tax valuation greater than the “fence” were removed, which totaled around 2,000 parcels. See figure 1 for a spatial distribution of this subset’s parcel tax valuation. This filtering process was only applied to the spatial autocorrelation portion of the analysis; another filtering method was used for the regression section, which is discussed in detail in the next section. We then defined the spatial relationships by 500ft (152 meter) distance. Based on this subset’s neighbor rule and spatial weights matrix, we computed the Global Moran’s I value, Global Moran’s I Scatterplot, Local Moran’s I values, and the LISA Cluster map.

Methods: Multilinear Regression

Due to the continued presence of extreme outliers and high leverage points, a final filter was used in R to only keep points with total value between \$100,000 and \$30,000,000. This was done using the following R code:

```
Tot_Val <- Tot_Val %>%  
  filter(TOTAL_VAL >= 100000 & TOTAL_VAL <= 30000000)
```

This filter retains 95.5% of the dataset while greatly improving the model fit—reviewed in the discussion portion of this report.

Prior to starting the regression analysis, it is important to identify any underlying correlation that might exist among variables in the dataset as well as the overall distribution of response values. These aspects are important for determining if any assumptions of linear regression are violated during the analysis.

A histogram of the Total Value in this dataset shows that the data is clearly and heavily right-skewed (see figure 2). We considered addressing this with a log transformation but

decided against it to maintain interpretability, as discussed further in the discussion. We used a scatterplot matrix (figure 3) to assess correlation among variables and a matrix of correlation to see numeric values of correlation.

To examine the relationship between tax valuation and brownfields, multilinear regression is an appropriate statistical method. Total Valuation (dollars) is the response variable, and list price (dollars), lot size (acres), number of RTNs within 500ft, duration days, inactive days, and contaminant type are the predictors. The formula based on these variables is as follows:

$$\text{Total tax valuation} = \beta_0 + \beta_1 \text{ List Price} + \beta_2 \text{ Lot Size} + \beta_3 \text{BFs within 500 ft} + \beta_4 \text{Duration} + \beta_5 \text{ Inactive} + \beta_6 \text{ OilYES} + \beta_7 \text{ Oil and HazMatYES} + \varepsilon$$

“Tot_Val_mod” was defined as the variable for the linear model fit. For clarity, the filtered dataset was defined as “Tot_Val” at the beginning of the analysis, and “TOTAL_VAL” is the variable name for the total tax valuation field in the dataset. The dummy variable for HazMat was not included because this variable was aliased (had perfect multicollinearity). The R code for this model fit is included:

```
Tot_Val_mod <- lm(TOTAL_VAL ~ LS_PRICE + LOT_SIZE + RTNs_Within_500ft +  
Duration_Days + Inactive_Days + Oil + Oil_and_HazMat, data = Tot_Val)  
  
summary(Tot_Val_mod)
```

The results of the regression model fit are summarized (figure 4) including the true coefficients and the Adjusted R-Squared values.

Discussion: Results and Diagnostics

The results of our spatial autocorrelation analysis give us context and insight as we make sense of the multi-linear regression results. We wanted to determine the impact that our predictors are having on parcel tax valuation and if variation in tax valuation is spatially autocorrelated. After running the Global Moran's I test in R, we found that Worcester parcel tax valuation has a low positive spatial autocorrelation value of 0.231 (see figure 5). This means that there is a low, albeit positive, tendency for parcels of high tax valuation to be located near other parcels with high tax valuation (HH), and parcels with low tax valuation to be near other parcels with low tax valuation (LL). The results of this test can be visualized in the Global Moran's Scatterplot, see figure 6, which has a positive slope and observation points mostly occupying the HH and LL quadrants. The LISA Cluster map also shows clear patterns of mostly HH and LL clustering, with HL areas being the apparent third most prevalent category of spatial autocorrelation. Specifically, the LISA Cluster map shows a large amount of HL areas along the southeast and south west areas of Worcester. LH areas are noticeable surrounding academic institutions, including Clark University (see figure 7). It is clear from these analyses that spatial autocorrelation is impacting the results of the multi-linear regression, and would need to be corrected using a spatially weighted regression.

Since we are only analyzing Worcester, and have a complete set of Worcester parcels and brownfields, we are working with a population dataset and therefore not concerned with hypothesis testing, p-values, t-values, or F-statistic value. The coefficients “estimates” as seen in the table are actually true coefficients.

As mentioned, this dataset has a severe right skew which is the first assumption violated in this linear regression analysis. Due to this fact, we considered using the log transformation to normalize the distribution, which did have a normalizing effect. However, this transformation had an adverse effect on the R-squared value. Since the goal of our analysis is to interpret the relationship between response and predictor variables, not prediction, this change was critical to our final decision. The log model ended up consistently having less than half the R-squared value of the regular model. This means that the regular model explains more than twice the amount of variability of total valuation than the log model. Additionally, the log model transforms the total valuation units (dollars) into log units (log dollars) which makes interpretation even more difficult. For these reasons, we decided that the regular model (un-transformed) was a better fit for this analysis.

There are several interesting points in the regression summary table (figure 4). The intercept value is $2.788e+05$, which means when all the predictor values = 0, Total Valuation = \$278,800. Simply put, for every one acre increase in Lot Size, Total Value increases by \$46,020; for every one RTN within 500ft of a parcel, Total Value increases by \$46,120; for every one oil contamination within 500ft of a parcel, Total Value increases by \$62,170; for every one Oil and HazMat (combination) contamination within 500ft of a parcel, Total Value increases by \$84,760; for every one day of RTN duration, Total Value decreases by \$18.23; for every one day an RTN has been certified or is otherwise no longer active, Total Value decreases \$5.13.

One question that stands out when reviewing these results is why do Oil and Oil/HazMat predictors have a positive relationship with total valuation? In his 2009 discussion paper, “The Effect of Voluntary Brownfields Programs on Nearby Property Values: Evidence from Illinois”, Joshua Linn discusses the concept of reverse causality regarding the propensity of property owners to report and certify brownfield sites:

“The benefit to the owner of certifying a brownfield increases with nearby property values. Sites located in areas with rising property values are therefore more likely to enter the program and receive certification. This raises the possibility of reverse causality, where rising property values cause certification...The effects of both entry and certification may occur with a lag of several years”

In other words, owners of higher value property are more inclined to go through the trouble of reporting and remediating contamination sites than lower value property owners. In addition, these property owners often improve conditions of their parcel during remediation to such a degree that property value increases as well. This might explain the positive relationship between these predictors and the response in the model.

Another question that also stands out is why does the inactive days predictor have a negative coefficient? Intuitively, this means that the more time an RTN within 500ft of a parcel has been inactive, the lower the property value (albeit by a very low rate). We have some thoughts on why this could be the case. It is possible that the definition of inactive days used in this model includes sites that have been given conditional certification for redevelopment/reuse while not completely remediating or removing the contaminant. Some of the Regulatory statuses may indicate that a minimum threshold of remediation was reached and progress then stagnated as further action was no longer legally mandated. In addition, RTN_Closed was included in this definition, however we know that this designation is often given to RTNs when they are consolidated with other RTNs at the same location for ease of tracking – meaning there is potentially still contamination present at the site. This might explain the negative relationship between inactive days and total valuation, though it remains uncertain.

The Residual Standard Error = \$1,043,000 which is high and indicates model predictions are typically off by \$1,043,000. This high value of RSE is likely inflated by outliers in the dataset, although the dataset is based on real estate values and is naturally going to be comparatively high. This result would be more concerning if the purpose of the analysis was to create a model with high predictive ability, but our goal is response/predictor relationship interpretation.

The Adjusted R-Squared = 0.1499, which means 14.99% of variability in Total Valuation can be explained by the model. The remaining 85.01% variability is explained by other variables not included in this model or by inherent noise/randomness in the dataset. However, we know from the spatial autocorrelation that this data is not completely spatially random. The low R-Squared value is likely caused by both missing variables and excessive noise in the dataset. This indicates that the model does not fit the dataset well.

A variety of diagnostic tests were also run to assess the model fit on this dataset. First, the VIF function was run on the model. No predictor got a value of over 2.2289 (Oil) which is considerably less than the problematic value of between 5-10 or greater. This indicates that there is no significant collinearity identified in this model. Second, the Spatial Pattern of Regression Residuals was mapped (see figure 8). Spatial correlation of the regression model residuals appears mild, though there are some visible clustering of high value residuals around university areas and clustering of low residual values near the center of town and along Main South. Lastly, the residual diagnostic plots were also assessed. The Residual VS Fitted plot (see figure 9) tests for non-linear relationships. Points are not in a cloud pattern or evenly distributed around 0 and the LOESS line has a clear upward slope which indicates a weak linear relationship. The Q-Q plot (see figure 10) checks if residuals are normally distributed. The LOESS line is almost completely flat around 0, instead of diagonal, which indicates a large portion of the dataset has small residuals. However, there are clearly a large number of outliers with extreme (mostly) high and low residuals causing heavy tails. This plot shows that the residuals are clearly not normally distributed. The Scale-Location plot (see figure 11) checks for constant variance (homoscedasticity). The unequal spread of points and a non-horizontal LOESS line suggest that the variance of residuals is not constant (there is heteroscedasticity) in this data. Additionally, there is a small check mark shape in the LOESS line toward the center of the cluster with the line extending out and up (likely due to outliers and high leverage points) as

fitted values increase and the points become more sparse. The Residuals VS Leverage plot (see figure 12) identifies influential observations (outliers and high leverage points) that affect the model fit. There are two unlabeled points that are nearly crossing Cook's .5 distance line above 0 and two labeled points that are well beyond Cook's 1.0 distance line below 0. The LOESS line is also being pulled up though there are no outliers or high leverage points in the direction of the line - we think this is due to the distribution of observation leverage values closer to 0 trending upward.

Overall, the assessment of this multilinear regression is based on the assumptions of linear regression that are violated or unviolated. Those assumptions are: 1) Linearity 2) Independence of Error 3) Constant Variance of Errors 4) Normality of Errors 5) No Perfect Multicollinearity. Among these assumptions, only the 5th is unviolated. The Residual VS Fitted plot appears to show weak linearity in this data, and data has heavy right skew (linear model is not a good fit). The Global Moran's I and LISA Cluster map confirm mild spatial autocorrelation of parcel Total Valuation values, and Spatial Pattern of Regression Residuals map confirms mild clustering within Total Valuation. There is correlation of error terms. The Scale-Location plot confirms that there is a significant level of heteroscedasticity (non-constant variance in residuals) in this dataset. The Q-Q Residuals plot shows heavy tails - errors are not normally distributed. Also, the Residuals vs. Leverage plot confirms outlier and high-leverage points remain in the dataset despite data processing/QC/filtering, though not as prevalent. Lastly, as confirmed by the VIF function, there is no collinearity in this dataset. The scatterplot matrix and the matrix of correlation also confirm that there are very low levels of correlation between predictors. It should be mentioned that the "HazMat" contamination type (dummy) variable was originally part of the model but removed due to perfect multicollinearity (aliased). See figure 13 for a summarized table of these assumption violations.

Conclusion

In all, this model does not sufficiently explain or predict the variability in total valuation and it violates all but one assumption of linear regression. There are likely other variables that can explain residual errors better. It is clear that multivariate regression alone is not sufficient to explain or predict the variability in total valuation.

Based on the results of our analysis, we see that brownfields have a complicated relationship with parcel tax valuation. Unexpected dynamics include the presence of contaminants, like oil, appearing to have a positive relationship with tax valuation, suggesting reverse causality in brownfield reporting and remediation tendencies. It is also clear that parcel tax valuation is spatially autocorrelated in Worcester, which likely impacted the results of our regression. Next steps for this project would include further dataset processing to better account for high levels of outliers, explore possibility of other brownfield related variables to include, and conduct a spatially weighted regression for this dataset to account for the spatial autocorrelation.

References

De Sousa, C., A., Wu, C., Westphal, L., M., 2009. Assessing the Effect of Publicly Assisted Brownfield Redevelopment on Surrounding Property Values. *Economic Development Quarterly*, 23 (2), 95-110.

Gillmor, P., E., 2001. *Small Business Liability Relief and Brownfields Revitalization Act* [online]. United States Environmental Protection Agency. Available from: <https://www.congress.gov/bill/107th-congress/house-bill/2869> [Accessed 04/26/2026].

Linn, J., 2012. The Effect of Voluntary Brownfields Programs on Nearby Property Values. *Resources for the Future*, Discussion Paper, 12-35.

Mihaescu, O., & vom Hofe, R. (2012). The impact of brownfields on residential property values in Cincinnati, Ohio: A spatial hedonic approach. *Journal of Regional Analysis & Policy*, 42(3).

Turečková, K., Martinát, S., Nevima, J., & Varadzin, F. (2022). The Impact of Brownfields on Residential Property Values in Post-Industrial Communities: A Study from the Eastern Part of the Czech Republic. *Land*, 11(6), 804. <https://doi.org/10.3390/land11060804>

Figures

Figure 1

Parcel Tax Valuation in Worcester, MA

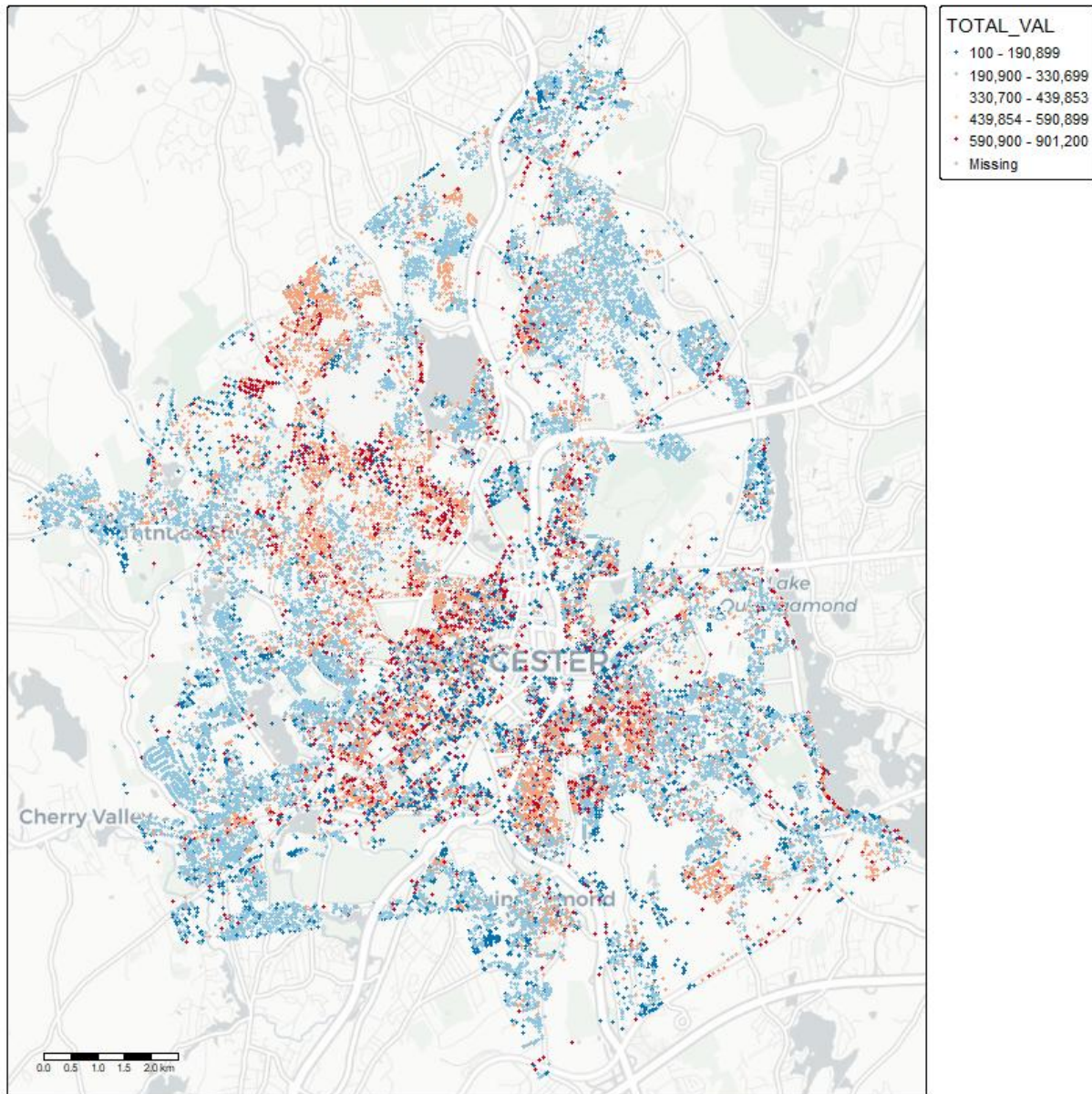


Figure 2

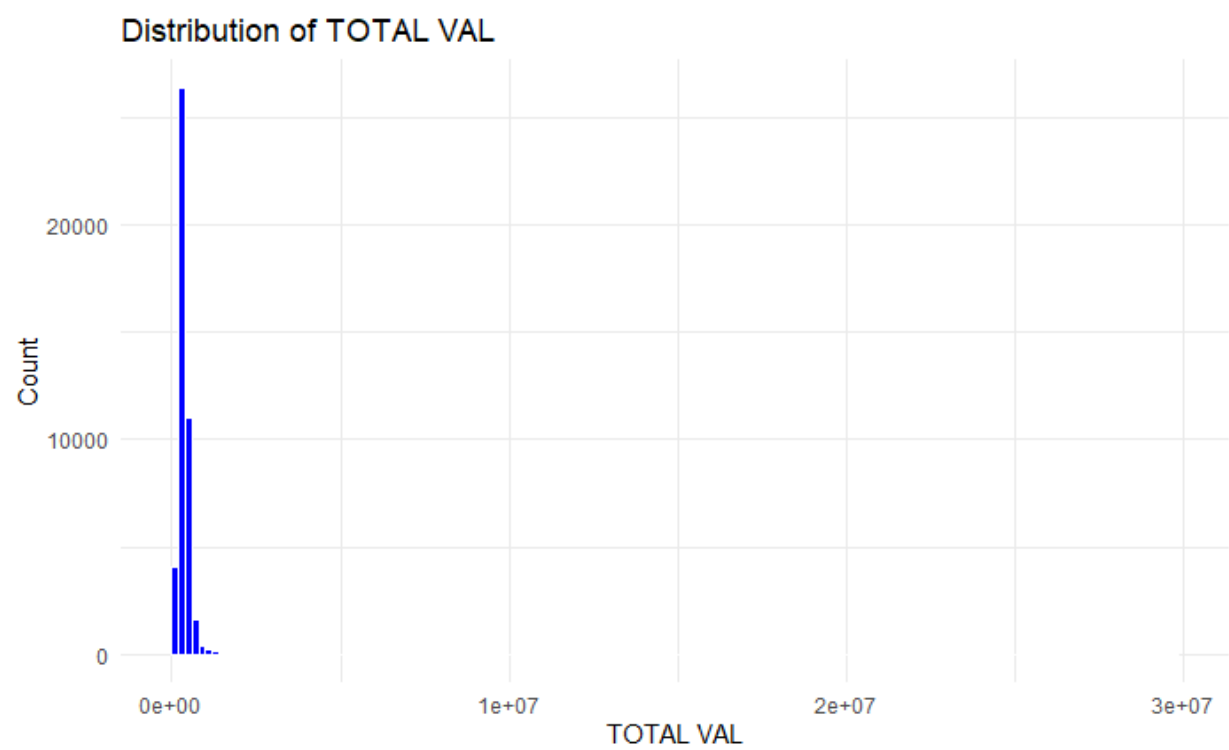


Figure 3

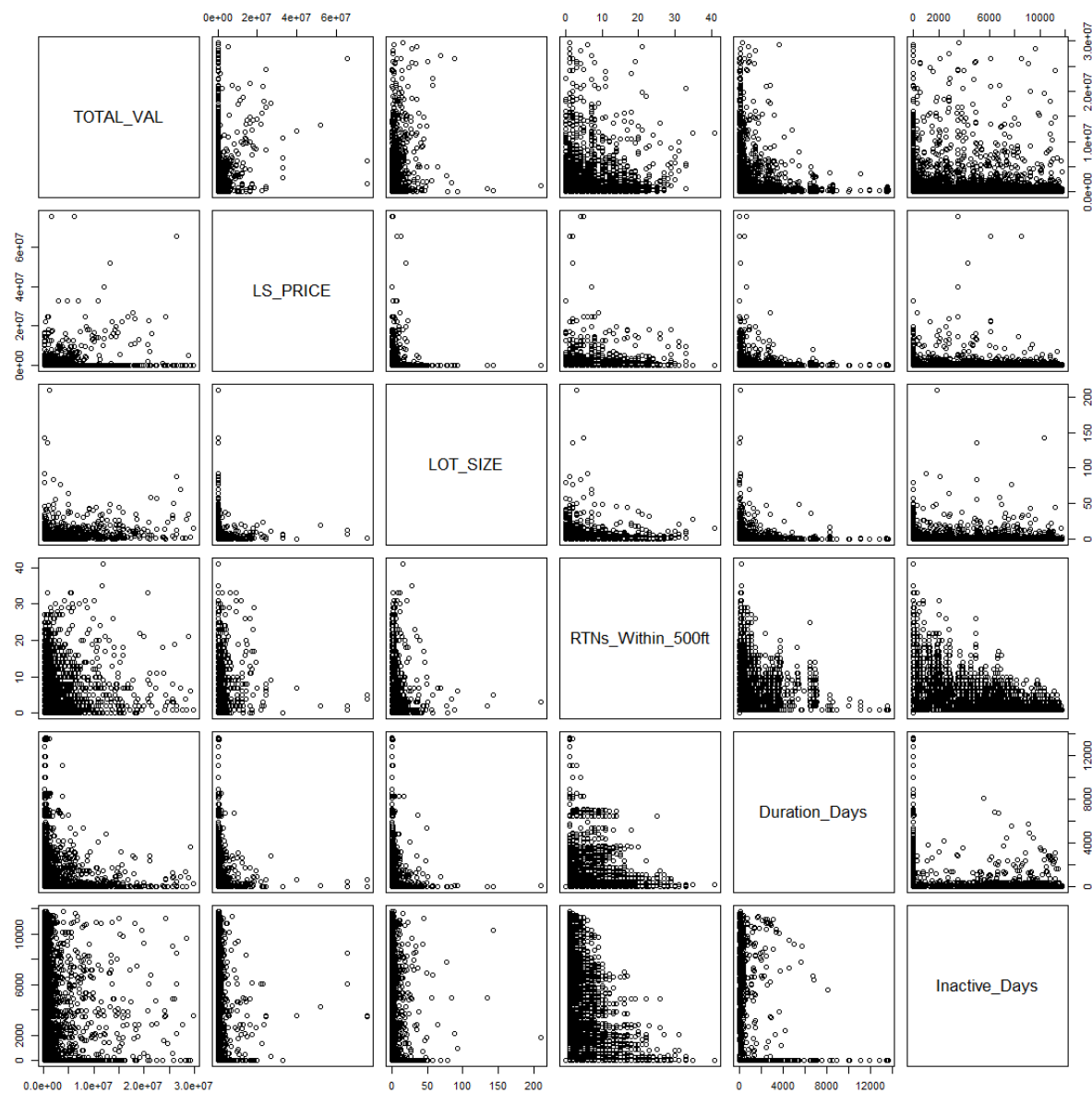


Figure 4

```
Tot_Val_mod <- lm(TOTAL_VAL ~ LS_PRICE + LOT_SIZE + RTNs_Within_500ft +  
Duration_Days + Inactive_Days + Oil + Oil_and_HazMat, data = Tot_Val) # HazMat removed  
because it is aliased
```

```
summary(Tot_Val_mod)
```

```
Call:
lm(formula = TOTAL_VAL ~ LS_PRICE + LOT_SIZE + RTNs_within_500ft +  
    Duration_Days + Inactive_Days + Oil + Oil_and_HazMat, data = Tot_val)

Residuals:
    Min       1Q   Median       3Q      Max
-21730038 -147326   -37922    46276  28671910

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  2.788e+05  6.935e+03  40.201  < 2e-16 ***
LS_PRICE      2.993e-01  4.383e-03  68.280  < 2e-16 ***
LOT_SIZE      4.602e+04  1.145e+03  40.190  < 2e-16 ***
RTNs_within_500ft 4.612e+04  1.793e+03  25.723  < 2e-16 ***
Duration_Days -1.823e+01  4.175e+00  -4.365  1.27e-05 ***
Inactive_Days -5.130e+00  1.876e+00  -2.734  0.00625 **
OilYES         6.217e+04  1.501e+04   4.141  3.47e-05 ***
Oil_and_HazMatYES 8.476e+04  2.586e+04   3.278  0.00105 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1043000 on 44875 degrees of freedom
Multiple R-squared:  0.15,    Adjusted R-squared:  0.1499
F-statistic: 1132 on 7 and 44875 DF,  p-value: < 2.2e-16
```

Figure 5

```
```{r, warning= FALSE, message= FALSE}
global Moran's I
moran.test(brownfield_sf$TOTAL_VAL, listw_dist)
```
```

Moran I test under randomisation

data: brownfield_sf\$TOTAL_VAL
weights: listw_dist
n reduced by no-neighbour observations

Moran I statistic standard deviate = 235.22, p-value < 2.2e-16
alternative hypothesis: greater
sample estimates:

| Moran I statistic | Expectation | Variance |
|-------------------|---------------|--------------|
| 2.311620e-01 | -2.207554e-05 | 9.660101e-07 |

Figure 6

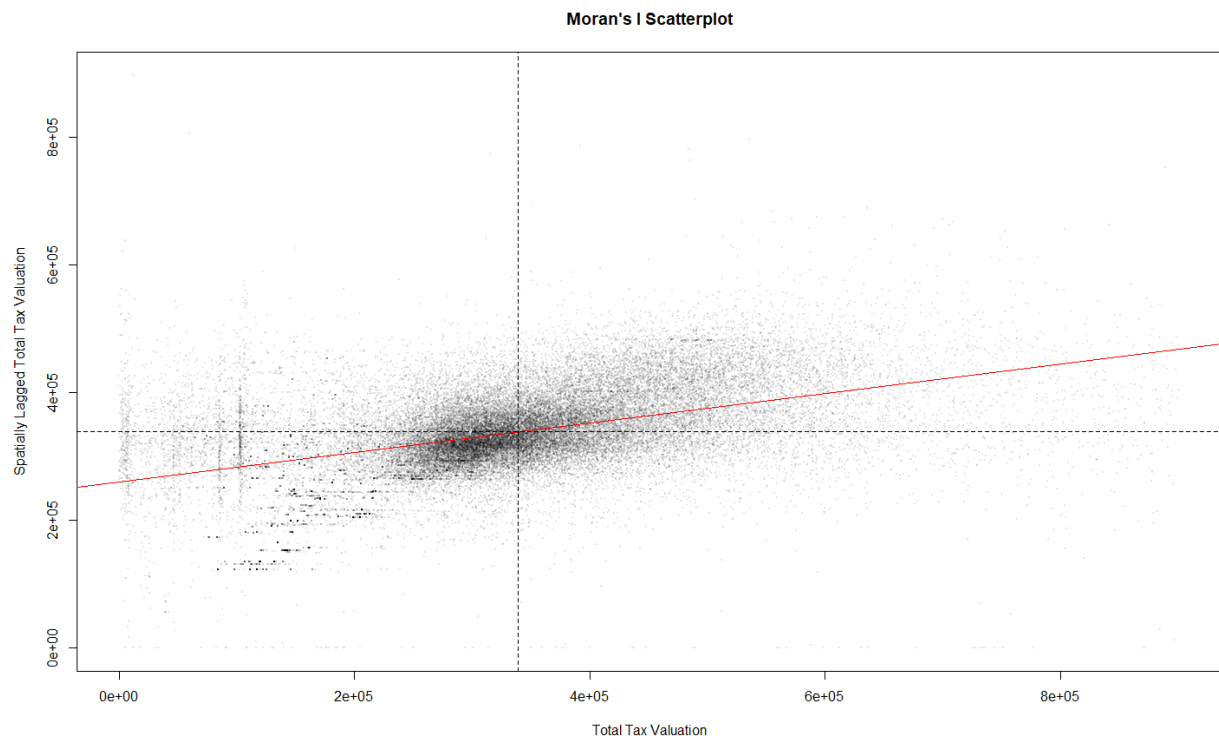


Figure 7

LISA Cluster Map

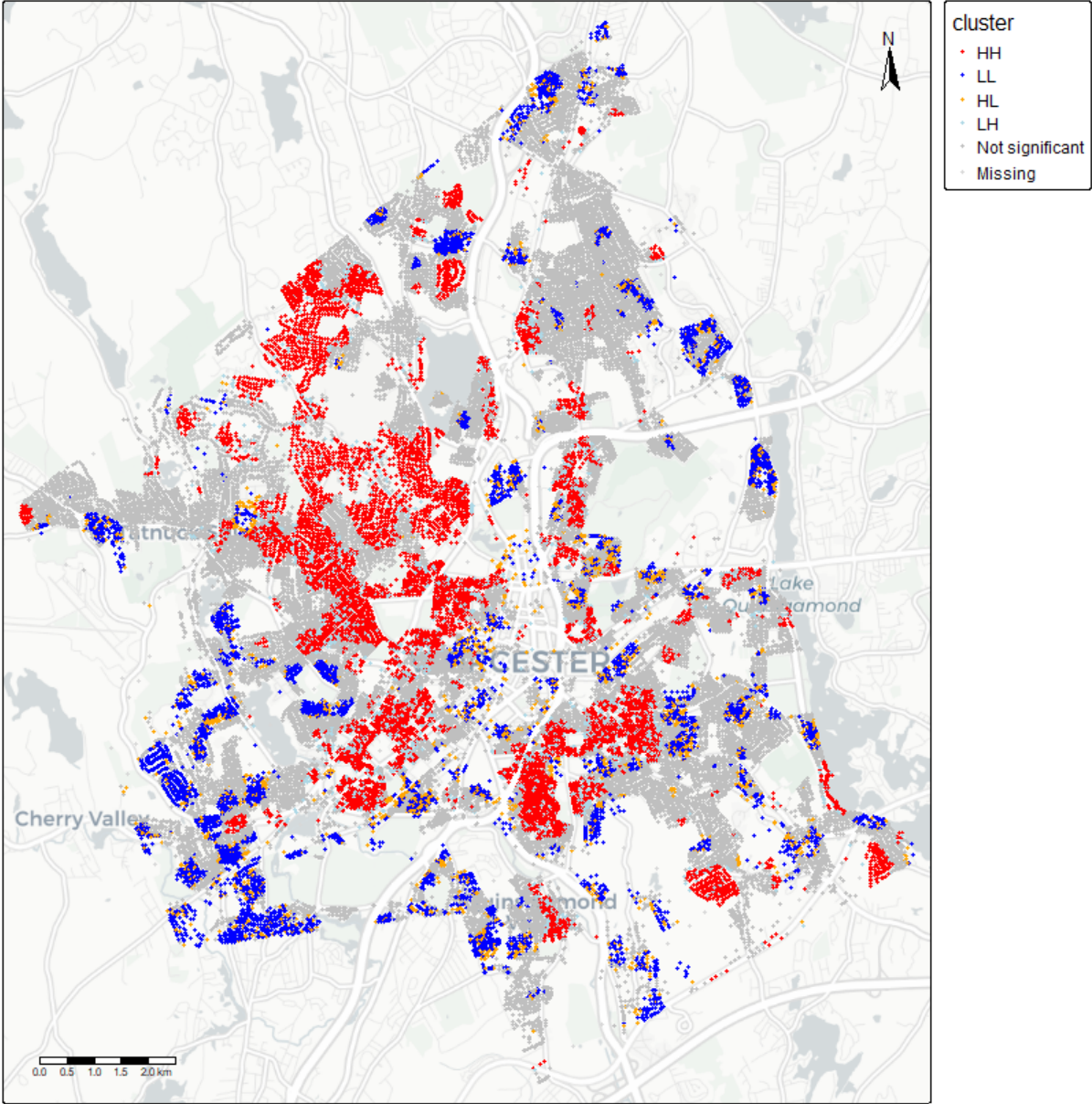


Figure 8

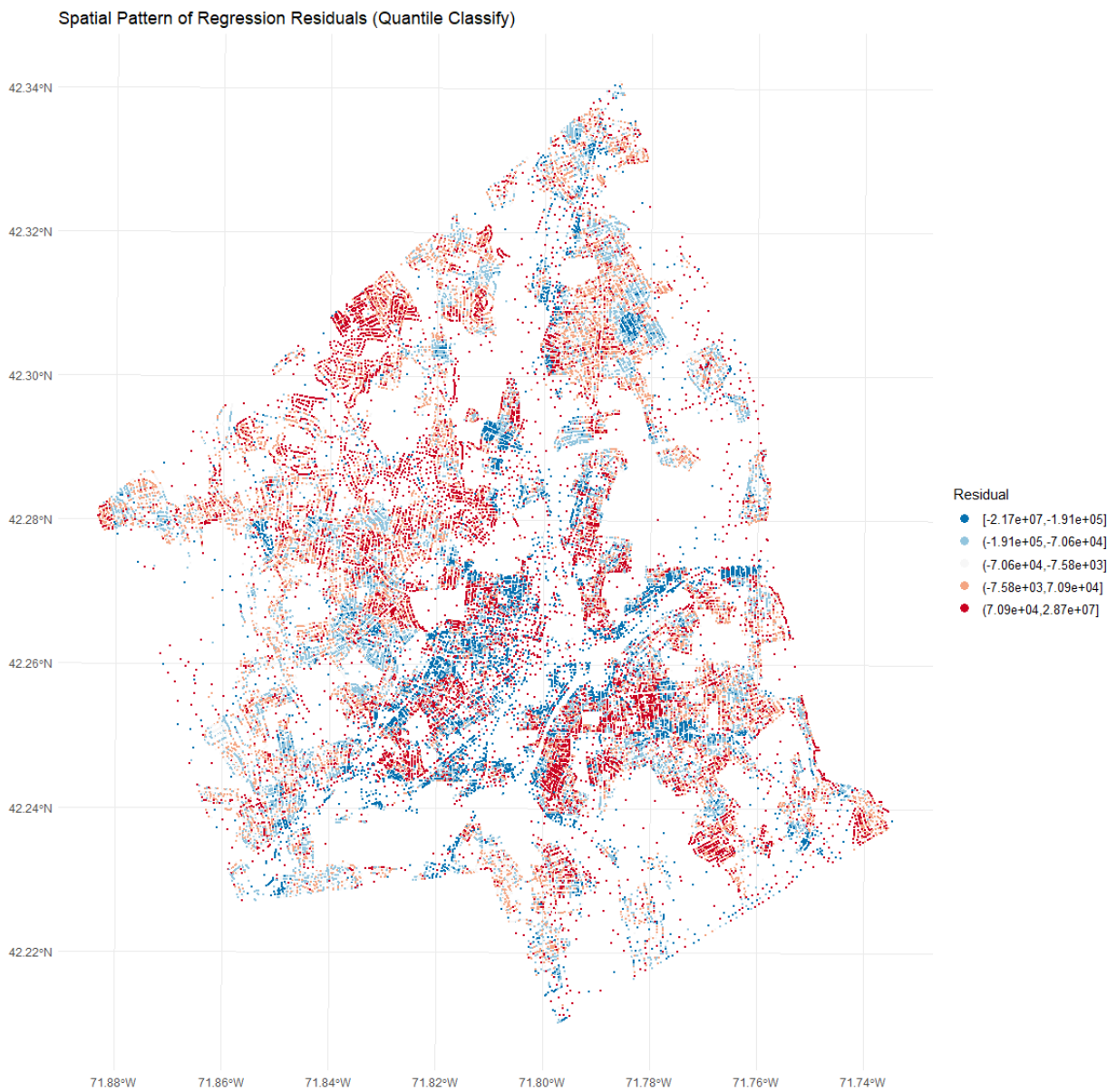


Figure 9

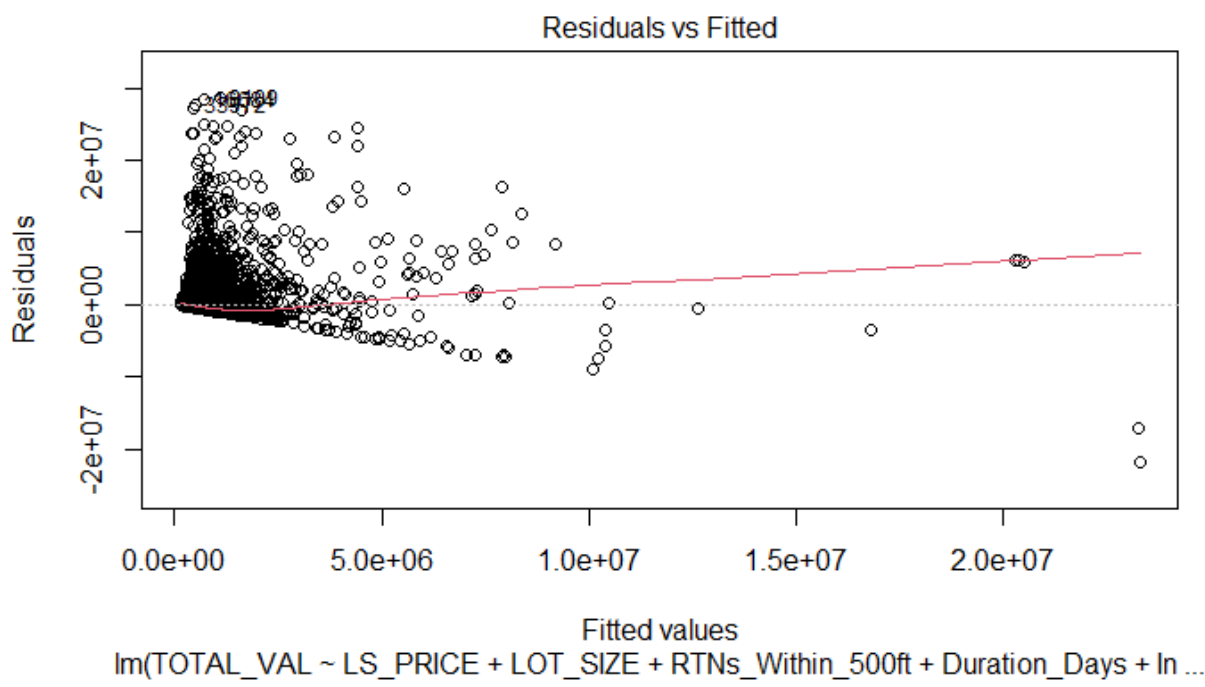


Figure 10

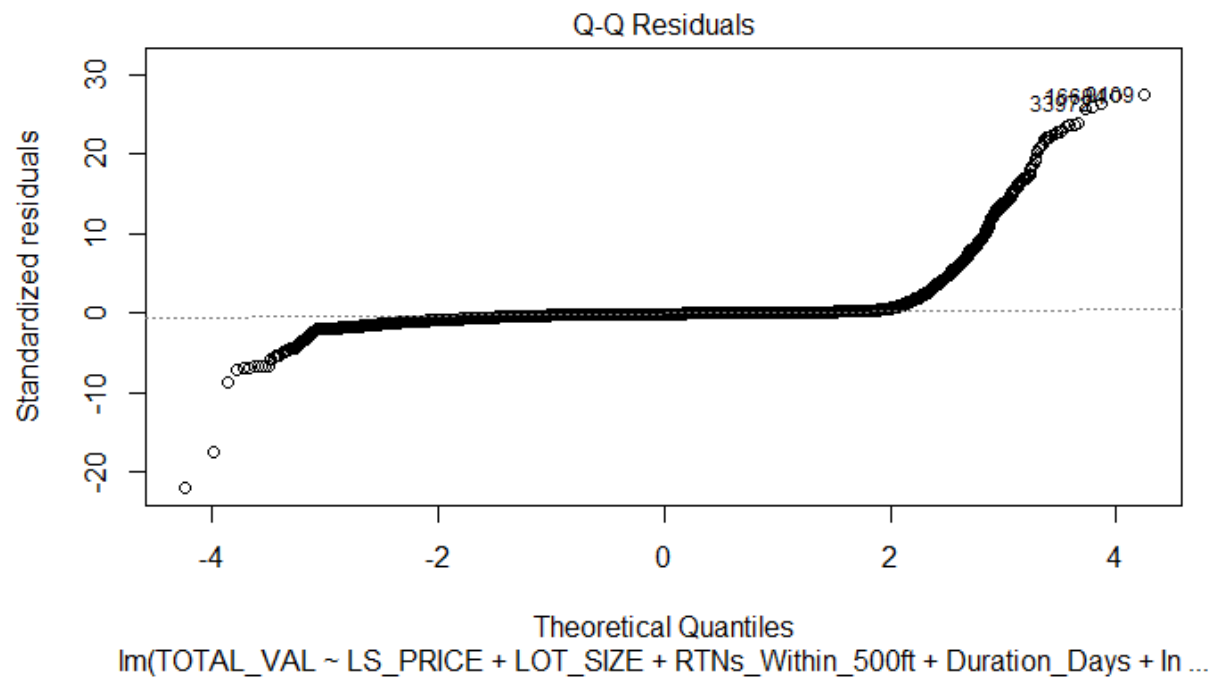


Figure 11

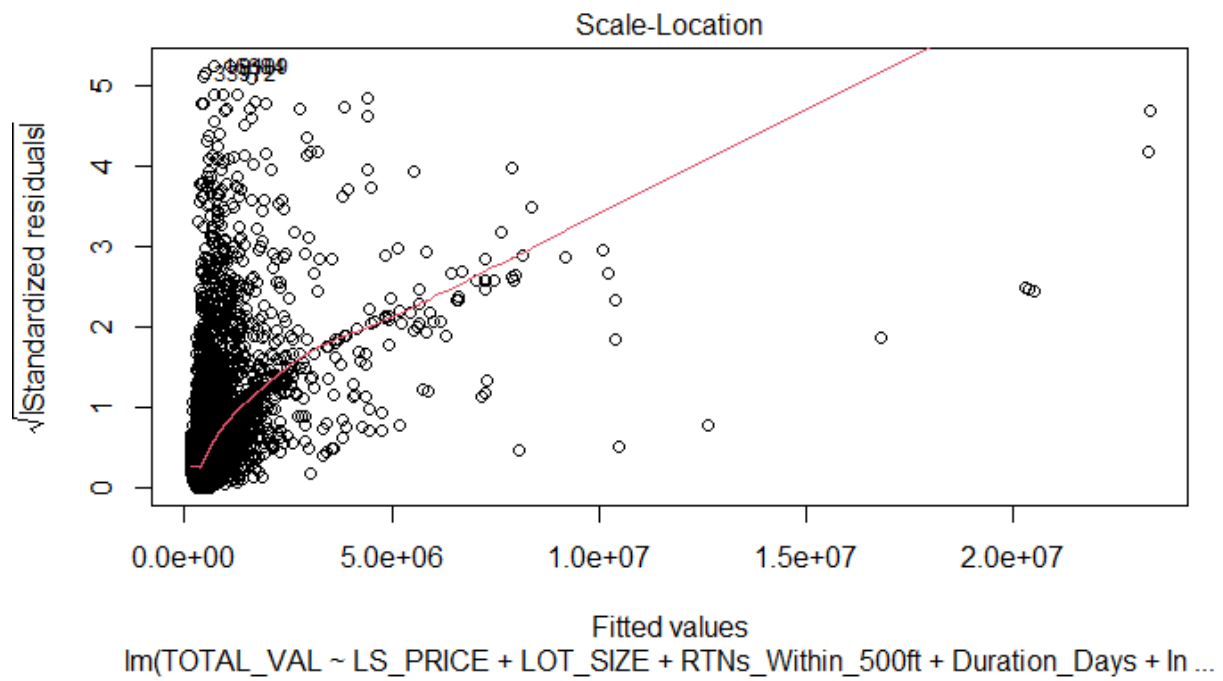


Figure 12

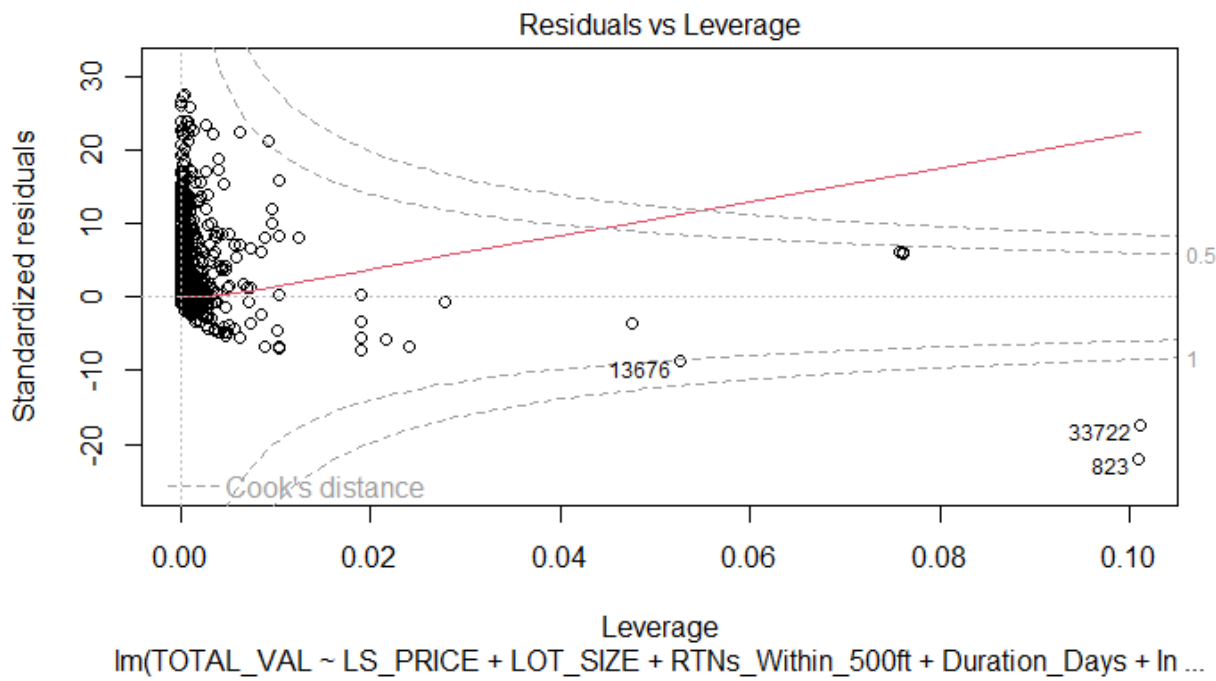


Figure 13

Assessment of Linear Regression Results

| | Assumption | Conform / Violate | Description |
|---|------------------------------|-------------------|---|
| 1 | Linearity | ✗ | <ul style="list-style-type: none"> Residual vs. fitted plot appears to show weak linearity Data is clearly right-skewed Linear model is not a good fit |
| 2 | Independence of Errors | ✗ | <ul style="list-style-type: none"> Correlation of error terms Global Moran's I and LISA Cluster map confirm mild spatial autocorrelation Spatial pattern of regression residuals confirms mild clustering |
| 3 | Constant Variance of Errors | ✗ | <ul style="list-style-type: none"> Non-constant variance of error terms Scale-Location plot confirm that there is a significant level of heteroscedasticity (non-constant variance in residuals) in this dataset |
| 4 | Normality of Errors | ✗ | <ul style="list-style-type: none"> Q-Q Residuals plot shows heavy tails - errors are not normally distributed Residuals vs. Leverage plot confirms outlier and high-leverage points remain in dataset despite data processing/QC/filtering, though not as prevalent |
| 5 | No Perfect Multicollinearity | ✓ | <ul style="list-style-type: none"> VIF function results show no value over 2.228956 Scatterplot matrix and matrix of correlation show low correlation between predictors No significant collinearity in this dataset (HazMat removed as aliased) |